A foundation model points to antigen presentation in SCLC T-cell dysfunction — and independent tissue agrees

Kaisar Dauyey · **Shinji Nakaoka**

Laboratory of Mathematical Biology, Faculty of Advanced Life Science, Hokkaido University

Model-derived dysfunction candidates in SCLC → Antigen-presentation programme → Associated with T-cell abundance in independent tumour tissue

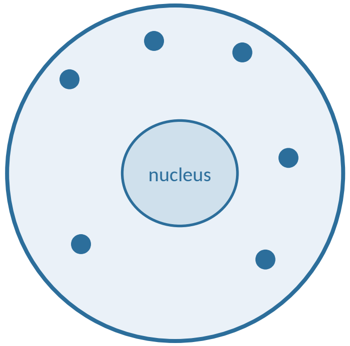
Hypothesis generator, tissue-validated — not a knockout screen

github.com/Kays3/geneformer-lung-tcell

What is an in silico perturbation?

Geneformer reads a cell as an **ordered list of its genes**, most-expressed first. Deleting or inserting one gene in that list and re-reading the cell asks a counterfactual question: *if this gene changed, would the cell look more like a different disease state?*

1 One T cell



2 Rank its genes

most expressed

- CD3D
- IL7R
- TIGIT
- GZMB
- TOX

least expressed

3 Edit one gene

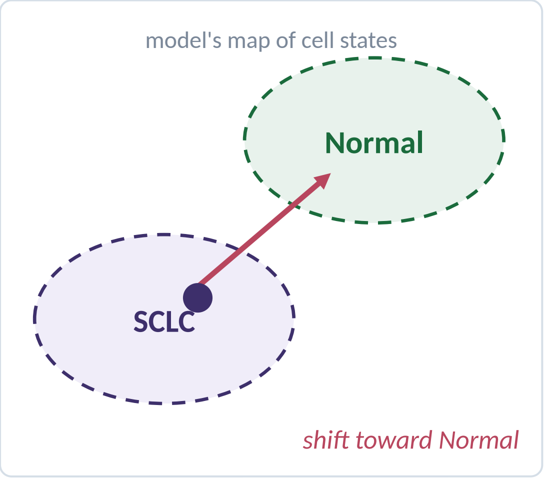
DELETE — drop it out

- CD3D
- IL7R
- ~~TIGIT~~ ✕
- GZMB ▲
- TOX ▲

the rest move up a rank

OVEREXPRESS — move it to the top

4 Measure the move



Nothing is edited in a laboratory here. The edit is made to the model's input and the result is the model's *predicted* change in cell state, averaged over thousands of real cells — a hypothesis generator, not a knockout experiment.

A gene counts only when deleting and overexpressing disagree

Every number in the next four slides is a perturbation shift. Two rules decide which ones are allowed to count.

CONCORDANT — counts as a hit

Delete the gene → toward Normal

Overexpress it → away from Normal

NOT CONCORDANT — discarded

Both edits move the cell the same way — the model is not responding to the direction of the change, so the gene is not treated as a candidate.

And it must replicate. The four headline edits hold the same sign in all three SCLC donors, at FDR < 0.05 in both arms, in genes detected in at least 100 source cells.

What the numbers are

A change in similarity

How much closer the perturbed cell sits to a target disease state, on the model's own scale.

Small by construction

Shifts run ~0.001–0.01. Size is not significance — slide 7 shows why.

Averaged over cells

Each value pools thousands of real T cells, not one simulated cell.

DATA PROVENANCE

42 individuals, zero donor leakage — so accuracy is measured on people the model never saw

Disease	Train	Eval	Test	Donors
SCLC	7,037	2,330	2,424	**19**
LUAD	17,831	5,611	6,387	**22**
Normal	2,334	1,620	566	**4**

45 donor×disease groups = 42 people + 3 paired tumour–normal donors

A donor contributing both tumour and normal tissue appears twice. The three extra rows are exactly RU675, RU682 and RU684.

46,140 cells · 24,540 features · median 390–1,172 cells per donor

Donor-disjoint splits

- **Train — 27 donors**
- **Eval — 7 donors**
- **Test — 8 donors**

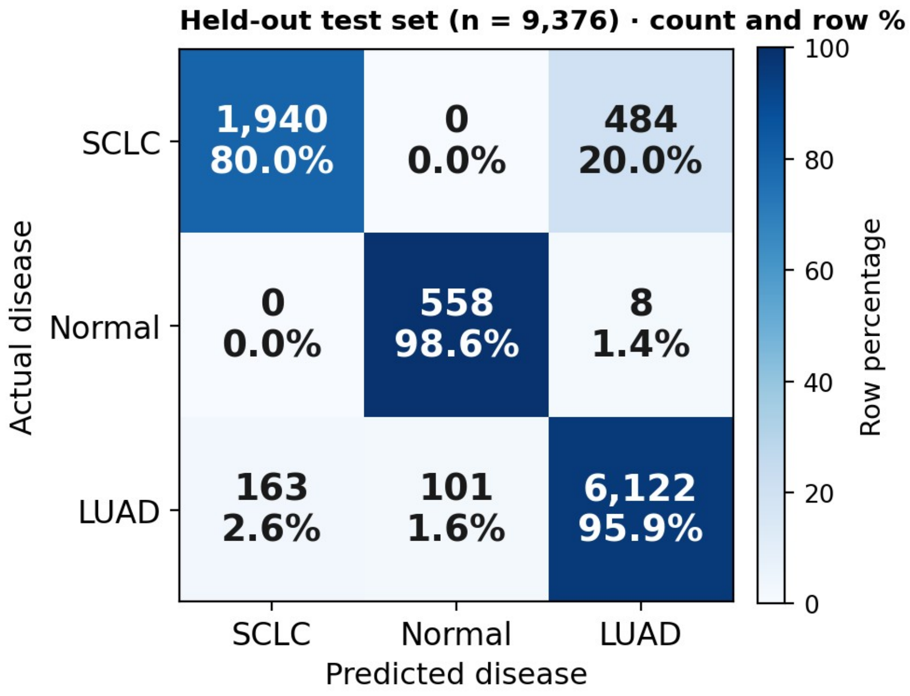
No individual appears in two splits. Paired tumour–normal donors are confined to one split by a cross-disease guard.

Held-out test set: 9,377 cells from 8 donors

Leakage audit: PASS · 0 / 42 donors

91.9% accuracy on held-out donors — with an SCLC→LUAD confusion worth noticing

91.9%	0.903	9,377	8
Test accuracy	Macro F1	Held-out cells	Test donors



Held-out test confusion matrix, row-normalised.

Look at the diagonal

- Normal 98.6% · LUAD 95.9%
- SCLC 80.0% — 484 missed cells go to LUAD
- Same axis as the open question on slide 7

Fine-tuning setup

- Geneformer-V2-104M
- 1 epoch · LR 5e-5 · 6 frozen layers
- NVIDIA GB10, 119 GiB unified memory
- ~48 h GPU · ~5.88M cell-gene perturbations

Four edits replicate across every donor: TIM-3, TIGIT, CTLA-4, IL7R

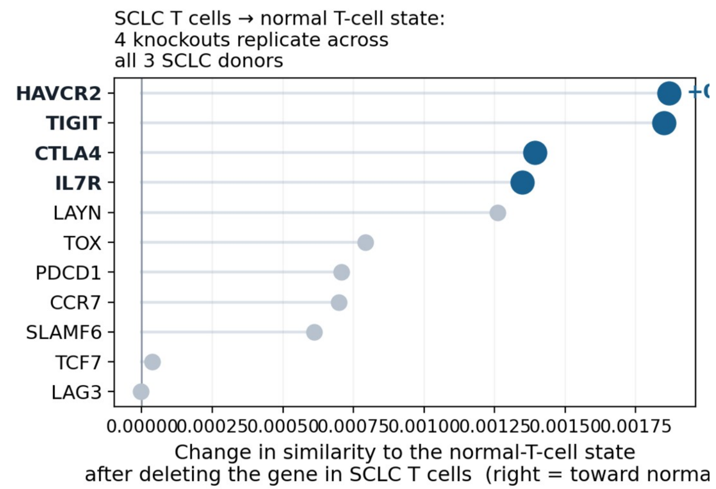
50	4	3/3	8
Genes screened	Replicated edits	Donors agree	No deletion result

Gene	Delete → Normal	Overexpress	Detected
HAVCR2 (TIM-3)	+0.0019	−0.0060	202
TIGIT	+0.0018	−0.0082	438
CTLA-4	+0.0014	−0.0014	282
IL7R (persistence)	+0.0013	−0.0054	1,131

All four: same sign in all 3 SCLC donors, FDR < 0.05 in both arms, ≥100 source cells detected.

50 genes › concordant both arms › FDR < 0.05 both arms › all 3 donors › ≥100 detected › **4**

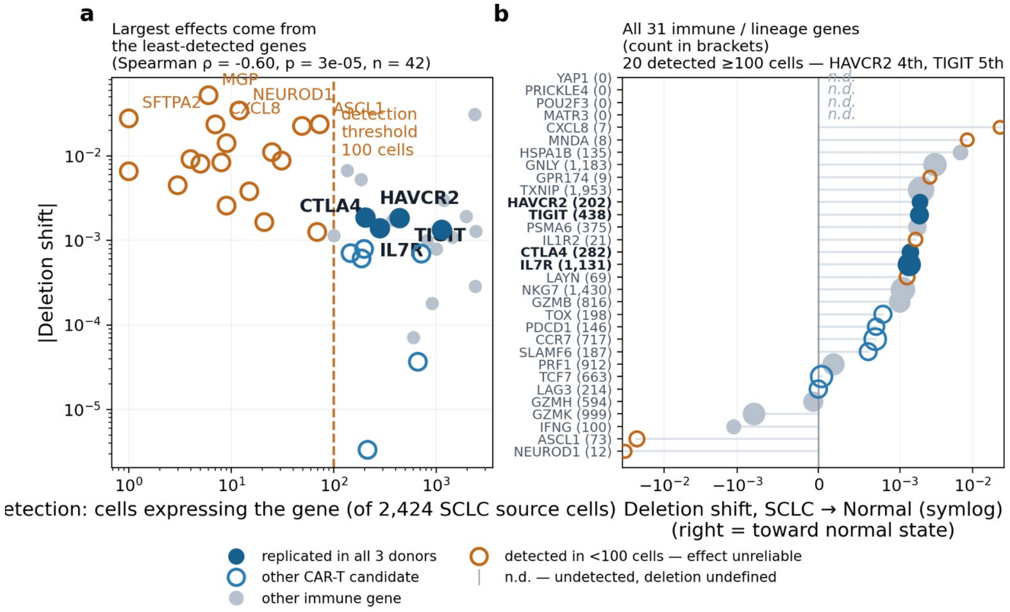
b



SCLC → normal: four knockouts replicate across all 3 donors.

QUALITY CONTROL

Sparse genes fake big effects (ρ = −0.60) — and one model result still contradicts the clinic

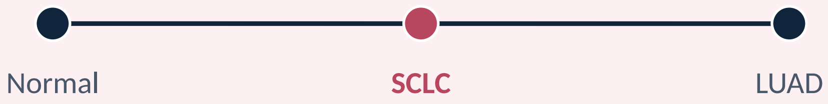


Detection vs absolute deletion effect across the full 50-gene panel.

- Detection vs effect size: Spearman ρ = −0.60 ($p = 2.7\times10^{-5}$, n = 42)
- 8 of 50 genes have no deletion result — undetected, so undefined, not zero
- 18 of 50 fall below 100 detected SCLC source cells
- Effect size alone is not evidence — which is why slide 6 ranked on replication

OPEN QUESTION

The model orders the exhaustion axis



TIGIT overexpression moves SCLC cells away from Normal (−0.0082) but toward LUAD (+0.0256).

Model axis: T-cell intrinsic

Clinic: tumour level — SCLC cold, ICI-resistant

Not the same measurement. Presented as an open question, not a reconciled result.

Two of the four hits are live CAR-T engineering targets — the tumour antigens are out of reach here

How interpretable is each hit?

TIGIT — Exhaustion checkpoint
Strongest external CAR-T support

TIM-3 (HAVCR2) — Exhaustion checkpoint
Direct KO precedent in solid tumours

CTLA-4 — Exhaustion checkpoint
Real signal, less CAR-T-specific evidence

IL7R — Persistence target
Persistence / fitness — not a checkpoint

This design CAN test

- T-cell-intrinsic checkpoint edits
- Persistence receptors (IL7R)
- Exhaustion-programme shifts in SCLC T cells

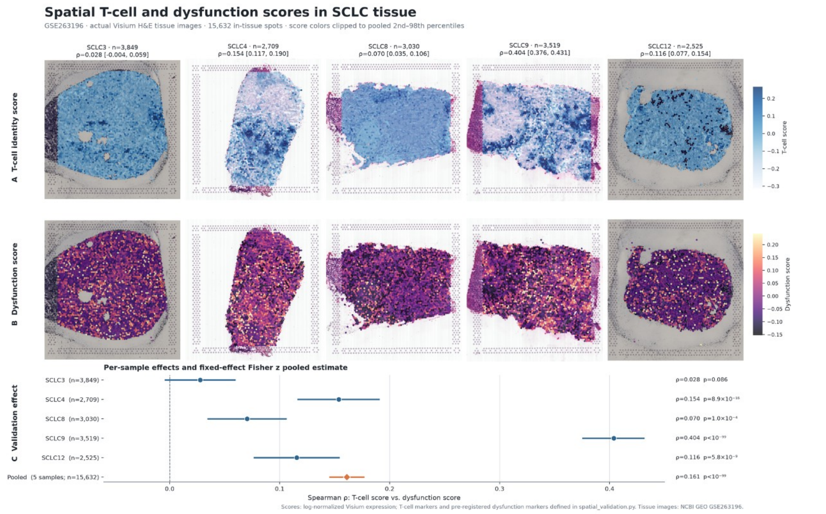
This design CANNOT test

- CAR-T tumour antigens: DLL3, SEZ6, NCAM1, CD276, CEACAM5
- PD-L1 (CD274) — absent from the T-cell atlas

Not expressed by T cells — this is scope, not failure. No PD-L1 claim is made.

In intact tumour tissue, dysfunction tracks T-cell abundance — and antigen presentation tracks it hardest

GSE263196 · 10x Visium · 5 specimens · 15,632 spots



A: T-cell identity. B: dysfunction. C: per-specimen and pooled ρ.

Why this is independent

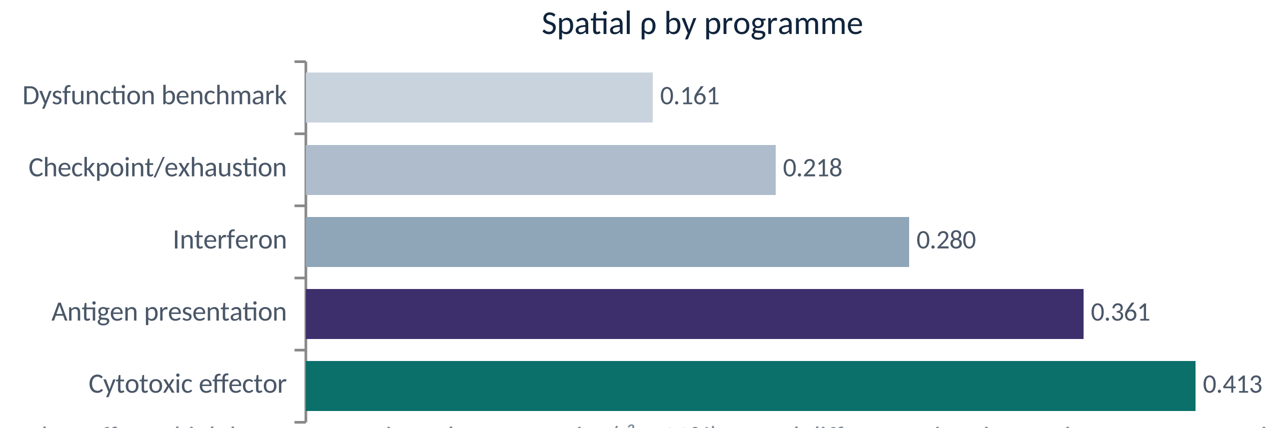
- Different cohort, assay and lab
- 15,632 / 15,774 in-tissue spots kept (99.1%)
- Marker panels share no genes — not a self-correlation

ρ = 0.161

7-gene dysfunction benchmark · [0.146, 0.176] · 4 of 5 specimens $p < 0.001$

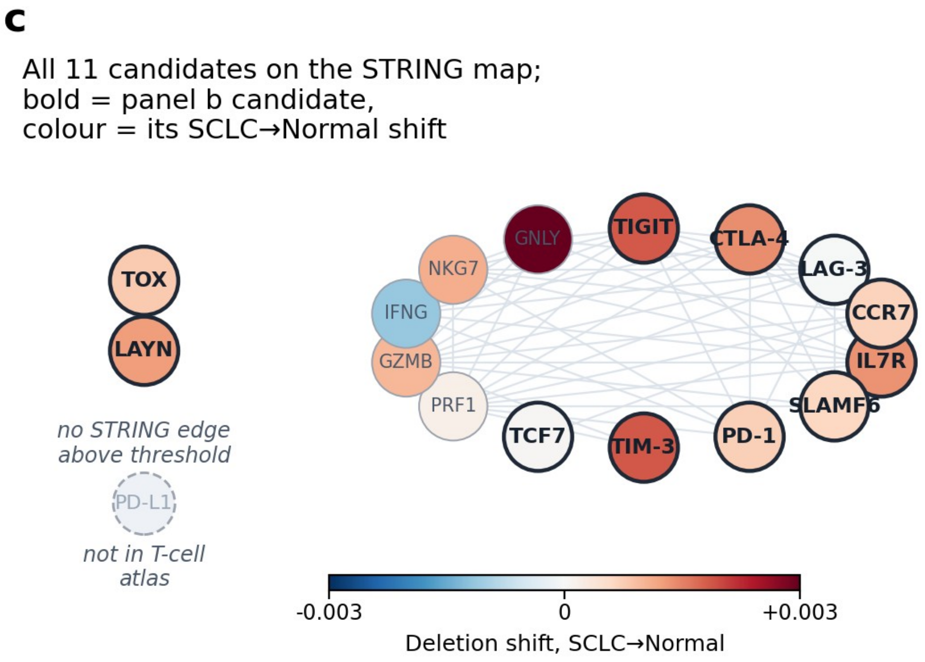
ρ = 0.361 → 0.384

13-gene antigen-presentation programme · depth-controlled · 7.95 σ above null



Modest effects, high between-specimen heterogeneity ($I^2 \approx 99\%$) — real differences in microenvironment composition, not measurement noise.

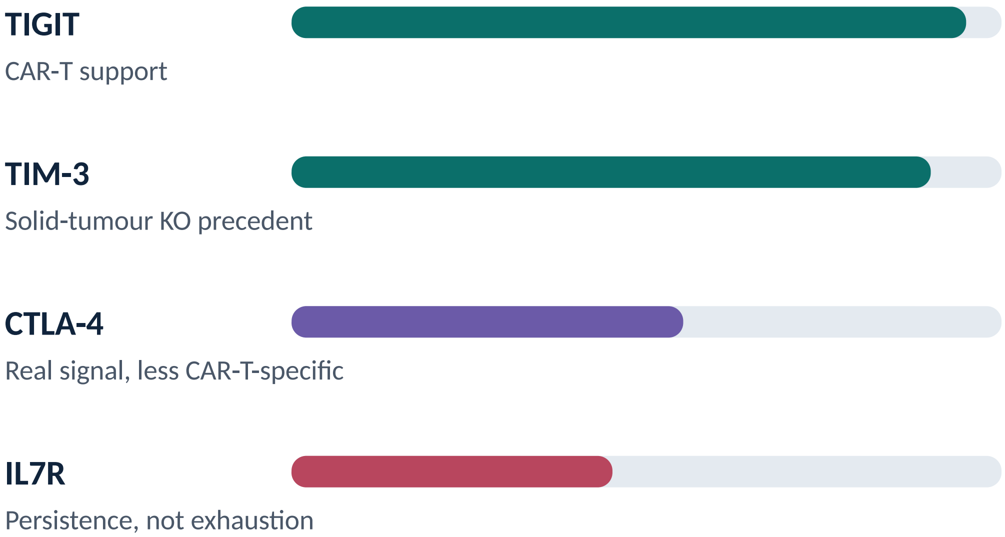
The hits sit in one connected interaction neighbourhood — from prior knowledge, not inferred here



All 11 ICI/CAR-T candidates on the STRING map; colour = SCLC→Normal deletion shift.

Prior interaction evidence — not a network inferred from these cells.

Evidence behind each hit



- 54 STRING edges among 16 context genes
- TOX, LAYN: no edge above threshold
- PD-L1 absent from the T-cell atlas

What this study does not claim

Model-derived predictions, not experimental knockouts

Perturbation shifts are interventions on a rank-value encoding. Everything upstream is a prioritised hypothesis.

Marker-score proxy, not deconvolution

Spot scores reflect panel expression, not inferred cell-type composition.

SCLC-specificity untested

The validation cohort has no Normal or LUAD sections to serve as controls.

Correlational, in archival tissue

Detection is source-state specific — absence of a result is not evidence of absence.

Stating these plainly is what makes everything on the previous ten slides usable.

A tissue-validated hypothesis, and the CRISPR experiment that would test it

✓ **Donor-clean model**
91.9% · F1 0.903

→

✓ **Four replicated edits**
TIM-3 · TIGIT · CTLA-4 · IL7R

→

✓ **Tissue-validated programme**
ρ 0.361 · 7.95 σ

EXPERIMENTAL

- CRISPR knockout of TIGIT and HAVCR2 in SCLC co-culture
- Perturb-seq in primary T cells vs in silico predictions
- Clinical outcome correlation once metadata is available

COMPUTATIONAL

- cell2location deconvolution in place of marker scores
- Normal / LUAD sections as specificity controls
- scGPT cross-model replication

Still open: why does the model order Normal < SCLC < LUAD on the exhaustion axis?

github.com/Kays3/geneformer-lung-tcell · snakaoka@sci.hokudai.ac.jp

Kaisar Dauyey & Shinji Nakaoka · Hokkaido University